Assignment Instructions

For each group, please submit the following documents:

1. **One PDF document** containing the answers to all questions below.
    - Use tables and figures where appropriate.
    - All tables and figures must be properly labeled and explained.
    - You must clearly explain **how each result is obtained** (methods, formulas, data treatment).
    - This document will be used for evaluation. **Lack of clear and precise explanations will result in a loss of points.**
2. **One Excel file** showing all calculations used to obtain your results.
3. **File naming convention**:
    - Name the PDF file **GX.pdf** and the Excel file **GX.xls(x)**, where **X** is your group number.
    - Example: Group 3 must submit **G3.pdf** and **G3.xls(x)**.

Part 1 – Chapters 2 and 3

For this part of the assignment, use the data provided in **Data_assignment.xls**. The file contains the following price series:

- **SP500**: price of the U.S. stock market index (labeled **SM**)
- **Nasdaq**: price of the U.S. Nasdaq index (labeled **N**)
- **Gold**: price of gold (labeled **G**)
- **Bitcoin**: Bitcoin price (labeled **B**)
- **Silver**: price of silver (labeled **S**)

In the following, **"the book"** refers to *Business Risk Management*.

Questions

1. **Weekly returns**
    - Compute **weekly returns** for each series.
    - Clearly state and use the same return definition throughout the assignment.
    - Report the **mean** and **standard deviation** of the weekly returns for each series.
2. **Scatter plots of returns**
   Create **four separate scatter plots** of weekly returns for the following pairs:
    - SM – N
    - SM – G
    - N – B

- G – S

When combining series that start at different dates, use **only the period in which both series are simultaneously available**.

3. **Empirical copula plots**
   Create **four empirical copula plots** of weekly returns for the same pairs:
   - SM – N
   - SM – G
   - N – B
   - G – S

   Construct the copulas following the procedure described in **Section 2.5 of the book** and in the lecture slides.
   When combining series with different starting dates, use **only overlapping observations**.

4. **Risk measures (weekly returns)**
   For **each of the five return series (SM, N, G, B, S)**:

   a) **Value at Risk (VaR)**

   - Compute the **empirical 95% and 99% VaR**, as in Section 3.2 of the book and Lecture 3 slides.
   - Assume returns are normally distributed with the mean and standard deviation computed in Question 1, and compute the **parametric (normal) 95% and 99% VaR**.

   b) **Expected Shortfall (ES)**

   - Compute the **empirical 95% and 99% ES**, as in Section 3.6 (page 83) of the book and Lecture 3 slides.
   - Using the same normality assumption as above, compute the **parametric (normal) 95% and 99% ES**.
5. **Monthly returns**
   - Repeat **all analyses in Questions 1–4** using **monthly returns** instead of weekly returns.

Part 2 – Chapters 4

Use the same data file **Data_assignment.xls** and the same five assets as in Part 1 (SM, N, G, B, S).
All questions in Part 2 are based on **weekly returns** r_t computed as in Part 1, and on **weekly losses** defined by

$$L_t = -r_t.$$

All tail analyses (tail index, GPD) must be performed on the **upper tail of losses** (large positive L_t).

1) Tail index (linear regression approach)

For each series (SM, N, G, B, S):

- Estimate the **tail index** using the **linear regression approach** described in Chapter 4 (Section 4.1.2), the lecture slides, and in class, applied to weekly losses L_t.
- Clearly state:
  - which observations are used (e.g., top k losses or exceedances above a threshold),
  - how k (or the threshold) is chosen, and why this choice is reasonable.
- Provide the **scatter plot used for the regression** and add the fitted regression line.
- Interpret the estimated coefficients (slope and intercept) in terms of tail behavior, according to the model used in the book/slides.
- Based on your estimated tail indexes, discuss which market appears **riskier in the tail** and explain why.

2) Generalized Pareto Distribution (GPD) estimation using the Expected Value property

For each series:

- Use the GPD expected value / mean excess property described in Chapter 4 to estimate the **two GPD parameters**(shape and scale) for the tail of weekly losses.
- Produce plots similar to Figure 4.10 of the book (e.g., mean excess plot across thresholds).
  In the PDF:
  - explain what is shown in the plot,
  - explain how the plot connects to the estimation method,
  - report the resulting parameter estimates.
- Interpret the results and compare them with the tail index conclusions from Question 1.

3) GPD estimation using Maximum Likelihood + tail risk measures

For each series:

- Fit a GPD to exceedances above a threshold u using **Maximum Likelihood (MLE)**.
- Choose u yourself and **justify** it (e.g., a sufficient number of exceedances).
- Report the fitted parameters and interpret them, also in light of Questions 1–2.
- Using the fitted GPD tail model, compute **unconditional** weekly **VaR** and **Expected Shortfall** at the **95%** and **99%** levels for losses.
- Compare the **empirical** 95% and 99% VaR/ES of losses to the GPD-based values (briefly explain similarities/differences and possible reasons).
- Detail each step (data preparation, threshold selection, estimation, formulas used).

Important (applies to every question in Part 2)

For every question, include a short but precise commentary explaining:

- the methodology used (including formulas),
- how the data were treated (weekly frequency, loss definition, thresholds / k, assumptions),
- and a brief interpretation of the results (and limitations, where relevant).

Important

For **each question**, provide a **detailed comment** explaining:

- the methodology used,
- how the data were treated (frequency, overlapping periods, assumptions),
- and a brief interpretation of the results.